

LARA TURGUT

Physics MSc Candidate, ETH Zurich | Machine Learning & Probabilistic Modeling

Location: Paul-Feyerabend-Hof 3, HWC 43, Zurich, Switzerland

Nationality: Turkish (B-permit in Switzerland)

Phone: +41 78 886 08 36

E-mail: laraturgut@hotmail.com

GitHub: github.com/lturgut



EDUCATION

Master Studies in Physics

ETH Zurich, Zurich, Switzerland

09/2023 – present

– GPA: 5.63/6.0

– **Master's Thesis (Grade: 6.0/6.0):** *Gauge-Equivariant U-Net Architecture for Diffusion Models and Continuous Normalizing Flows*

Implemented and evaluated a gauge-equivariant U-Net architecture for generative modeling within an international research collaboration, applying diffusion models as efficient alternatives to traditional Markov Chain Monte Carlo sampling methods for complex probability distributions.

– **Master's Research Project:** *Invertible U-Net Architecture for Scalar Lattice Field Theory: A Renormalization Group Perspective*

Designed an invertible U-Net architecture within a normalizing flow framework for sampling complex probability distributions, evaluating efficiency using effective sample size.

– **Selected coursework:** Machine Learning for Finance and Complex Systems, Information Systems for Engineers, Principles of Microeconomics, Principles of Macroeconomics.

Master Studies in Physics, Exchange Semester

EPFL, Lausanne, Switzerland

09/2023 – 02/2024

Physics BSc

ETH Zurich, Zurich, Switzerland

09/2019 – 09/2023

– **Bachelor Project:** *The Caesium Magnetometer Array of the Gradiometer for the n2EDM Experiment at PSI.*

Highschool degree (Abitur)

Istanbul High School, Istanbul, Türkiye

09/2014 – 06/2019

– Top German international high school in Türkiye with science-mathematics orientation.

– Ranked in the top 0.04% in the national high school entrance examination of Türkiye.

PRACTICAL EXPERIENCE

Teaching Assistant, Numerical Methods

ETH Zurich, Zurich, Switzerland

02/2024 – 05/2024

– Conducted exercise sessions covering numerical ODE solvers (Runge-Kutta, implicit, symplectic, and stiff methods), numerical linear algebra, and iterative methods for nonlinear equations (Newton, Broyden).

Internship, Isotope Physics Research Group

University of Vienna, Vienna, Austria

07/2023 – 08/2023

– Built a Python pipeline for data evaluation of actinide measurements using accelerator mass spectrometry.

Teaching Assistant, Physics II

ETH Zurich, Zurich, Switzerland

02/2023 – 06/2023

Teaching Assistant, Linear Algebra

ETH Zurich, Zurich, Switzerland

09/2021 – 12/2021

IT-SKILLS

Programming languages: Python (NumPy, SciPy, Pandas, Matplotlib), C++, Julia.

Machine learning: PyTorch, Scikit-learn.

Database management: PostgreSQL.

Version control: Git.

Other computer skills: LaTeX, Docker

PROJECTS

LLM Benchmarking for Financial Reasoning with Apertus

03/2026 – present

ETH Zurich, Zurich, Switzerland

- In collaboration with Apertus (Swiss LLM initiative).
- Benchmarking large language models on compositional financial reasoning tasks in an interdisciplinary team.
- Designed and generated synthetic financial datasets and multi-step Q&A pairs using expression trees to evaluate model reasoning.

Berezinskii–Kosterlitz–Thouless Transition in the 2D XY Model

04/2025 – 05/2025

ETH Zurich, Zurich, Switzerland

- Collaborated in an interdisciplinary team to simulate the 2D classical XY model using the Wolff cluster algorithm.

COMPETITIONS

1st Place (Team), UN OCHA Datathon Challenge

04/2026

- Challenge posed by the United Nations Office for the Coordination of Humanitarian Affairs (UN OCHA) at Switzerland’s largest ML hackathon (Datathon).
- Built an explainable Bayesian model to identify under-funded humanitarian crises through a latent variable capturing systematic donor neglect, informed by needs, funding, and donor priorities.

1st Place (Team), UBS Case Competition at ETH Zurich

03/2026

- Asset management & investment banking simulation.
- Referred directly to UBS recruitment based on individual performance.

PUBLICATIONS

Diffusion Models for $SU(N)$ Gauge Theories

05/2026

J. Komijani, M. K. Marinković, **L. Turgut**

arxiv.org/abs/2605.06134

EXTRACURRICULAR EXPERIENCE

Board Member & Head of Marketing, Analytics Club at ETH (ACE)

09/2025 – present

ETH Zurich, Zurich, Switzerland

- Led marketing strategy and event promotion for one of Europe’s largest student-run data science societies.

Marketing Committee Member, ACE

03/2025 – 09/2025

ETH Zurich, Zurich, Switzerland

Board Member & Public Relations, MindPhair

10/2023 – 10/2024

ETH Zurich, Zurich, Switzerland

- Organized outreach and communication for a career fair targeting students in mathematics, physics, and computational sciences.

Board Member & Treasurer, VMP

02/2023 – 09/2023

ETH Zurich, Zurich, Switzerland

- Managed budgeting, financial reporting, and payment operations for VMP, the Mathematics, Physics and CSE student association at ETH Zurich.

LANGUAGES

English (Fluent), German (C1 – DSD II), Turkish (Native)

HOBBIES

Skiing, running, gym.